

TradeFlow v2 API Reference

TradeFlow v2 — Complete API & Function Reference

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Stack: Next.js 16, React 19, Supabase, NVIDIA NIM, Resend, Stripe, WhatsApp Cloud API

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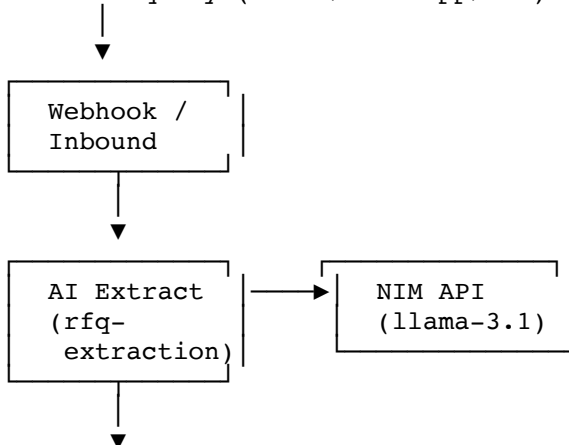
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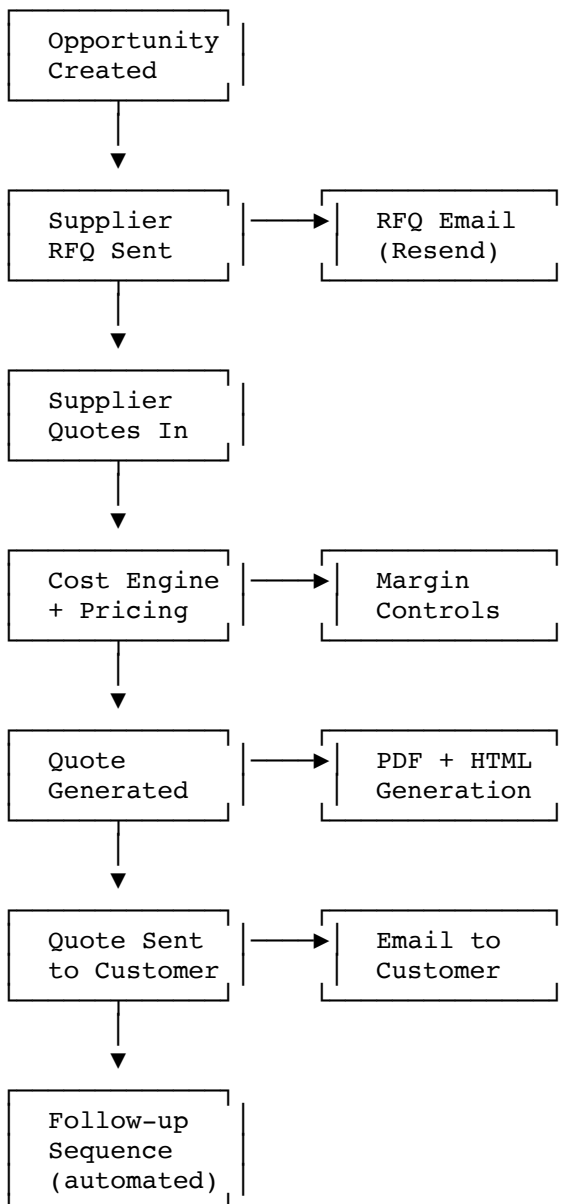
1. Overview

TradeFlow v2 is an AI-powered inquiry-to-quote trading operations platform for HK trading/sourcing companies. The system processes incoming customer inquiries (via email, WhatsApp, or web form), extracts structured requirements using AI, generates supplier RFQs, compares supplier quotes, builds customer-facing quotes with margin controls, and automates follow-up sequences.

Architecture

Customer Inquiry (email/WhatsApp/web)





2. Authentication & Authorization

src/lib/api-auth.ts — Server-Side Auth

requireAuth(req, options?): Promise<AuthResult>

Extracts and verifies the Supabase JWT from the Authorization header or cookies. Returns user + company info. Throws a Response (401/403) on failure.

- **Input:** NextRequest, optional { requireCompany?: boolean }
- **Output:** { user: { id, email }, companyId: string | null }
- **Auth flow:** Extracts Bearer token → supabaseAdmin.auth.getUser() → queries users table for company_id
- **Error cases:** Missing token (401), invalid token (401), user not found (401), no company (403)

src/lib/auth-fetch.ts — Client-Side Auth

authFetch(url, options?): Promise<Response>

Client-side fetch wrapper that automatically attaches the Supabase auth token from the browser session. Uses 'use client' directive.

- **Input:** URL string, optional RequestInit
- **Output:** Standard Response
- **Dependency:** supabaseBrowser from @/lib/auth

src/lib/rate-limit.ts — Rate Limiting

checkRateLimit(key, config?): { allowed, remaining, resetTime }

In-memory sliding window rate limiter. Per-process only (not distributed across instances).

- **Default:** 100 requests per 60-second window
- **Output:** { allowed: boolean, remaining: number, resetTime: number }

getClientIp(req): string

Extracts client IP from x-forwarded-for header or falls back to 'unknown'.

createRateLimitResponse(resetTime): Response

Returns a 429 JSON response with Retry-After header.

3. Library Modules

src/lib/supabase.ts — Database Clients

Exports two Supabase clients:

- **supabase** — Browser client with anon key (auto-refresh, session persistence)
- **supabaseAdmin** — Server client with service role key (no auto-refresh, no RLS bypass via client, but full admin access via service role)

Env vars: NEXT_PUBLIC_SUPABASE_URL, NEXT_PUBLIC_SUPABASE_ANON_KEY, SUPABASE_SERVICE_ROLE_KEY

src/lib/ai.ts — Core AI Chat Handler

Handles incoming customer conversations via WhatsApp/WeChat. Builds system prompts from company data, calls NVIDIA NIM API, and manages conversation state.

handleChat(ctx: ChatContext): Promise<ChatResponse>

Main chat handler. Loads company products, FAQ rules, and knowledge base. Detects language, matches FAQ patterns, falls back to NIM AI, and saves messages to the database.

- **Input:** { companyId, channel, conversationId, userMessage, contactPhone, contactName }

- **Output:** { reply: string, tokensUsed: number }
- **Flow:** Load company context → detect language → match FAQ → build system prompt → call NIM → save messages
- **NIM config:** meta/llama-3.1-8b-instruct via <https://integrate.api.nvidia.com/v1>

checkKeywordOverride(message): boolean

Detects human handoff keywords: “human”, “真人”, “speak to person”, “manager”, etc.

generateHandoffSummary(conversationId, companyName): Promise<string>

Generates a 2-line summary of a conversation for human takeover context.

getOrCreateConversation(companyId, channel, contactIdentifier, contactName?): Promise<string>

Finds or creates an active conversation for a contact on a given channel.

src/lib/rfq-extraction.ts — AI Requirement Extraction

Extracts structured trading request data from email text + attachments using NIM API. Provides field-level confidence tracking.

extractTradingRequest(params: ExtractionInput): Promise<TradingRequest>

Main extraction function. Converts free-form email text into a structured TradingRequest with CONFIRMED/EXTRACTED/INFERRED/MISSING/CONFLICTING status per field.

- **Input:** { emailText, attachmentTexts[], companyProducts[], requirementTemplate?, subject?, sourceChannel? }
- **Output:** TradingRequest with customer info, product requirements, delivery terms, commercial terms, and confidence metadata
- **Fields extracted:** Customer name/email/company, product name/description/specs, quantity/unit, target price, delivery date, destination, incoterms, payment terms

identifyMissingFields(extracted, template): MissingField[]

Compares extracted data against a product requirement template. Returns missing required/recommended fields with human-friendly suggestions.

draftClarificationEmail(inquiry, missingFields): Promise<string>

AI-drafted email asking the customer for missing information.

classifyInquiryType(text): Promise<string>

Classifies inquiry as: RFQ, PO, inquiry, complaint, technical_question, or general.

src/lib/cost-engine.ts — Cost & Margin Calculator

Deterministic, stateless cost and margin calculation engine. Pure arithmetic — no LLM calls.

calculateCostBuildUp(params: CostBuildUpInput): CostBuildUpResult

Aggregates 9 cost components (product cost, tooling, packaging, inspection, transport, freight, insurance, financing, contingency) into a total estimated cost with line items.

calculateCustomerPrice(costBuildUp, marginRule): CustomerPriceResult

Derives customer-facing price from cost + margin rule (fixed amount, percentage, or tiered).

validateCostBuildUp(costBuildUp, companySettings): CostValidationResult

Validates against company policy: missing components, expired quotes, currency mismatches, negative margins.

convertCurrency(amount, from, to, rates): number

Currency conversion using flat rate table.

roundQuoteAmount(amount, currency): number

Smart rounding (0 decimals for JPY/KRW/VND, 2 for others).

src/lib/quote-generator.ts — Quote Generation Engine

Full quote lifecycle: builds quotes from opportunities, generates PDF/HTML, handles versioning and pre-send validation.

generateQuoteNumber(companyId): Promise<string>

Sequential quote number in QT-YYYY-NNNN format. Uses database-level lock to prevent race conditions. Falls back to timestamp-based number.

buildQuoteFromOpportunity(params: QuoteBuildInput): Promise<QuoteBuildResult>

Constructs a draft Quote from opportunity data, selected supplier quotes, and cost build-up. Returns hydrated draft with line items and cost components.

generateQuotePDF(quote, company, lineItems, options?): Promise<Buffer>

Generates a branded PDF quotation document. Pure PDF object construction (no external dependencies). Includes company header, pricing table, terms, and footer.

- **Options:** includeLogo, includeCostBreakdown, includeTerms, customNotes
- **Output:** PDF document as Node.js Buffer

generateQuoteHTML(quote, company, lineItems): string

Generates a responsive HTML email body for sending the quote. Works across email clients. Includes summary table and call-to-action button.

createQuoteVersion(quoteId, companyId, changeSummary): Promise<void>

Snapshots current quote state as a new version record. Never overwrites or modifies sent quotes.

validateQuoteForSending(quote, costBuildUp, companySettings): ValidationResult

Validates: required fields, margin thresholds, cost completeness, validity dates, and terms inclusion. Returns issues with severity (error/warning).

src/lib/follow-up-engine.ts — Follow-up Automation

Manages scheduled follow-up sequences. Uses NIM API for contextual message generation — never sends generic “just following up” messages.

createDefaultSequence(params: SequenceInput): Promise<FollowUpSequence>

Creates a 4-step follow-up sequence: - Day 3: Check-in (references specific product/quote details) - Day 7: Needs update (asks if requirements changed) - Day 14: Value add (offers revised options/samples) - Day 21: Close (polite close, leaves door open)

pauseSequence(sequenceId, companyId, userId): Promise<void>

Pauses all scheduled items in a sequence.

resumeSequence(sequenceId, companyId): Promise<void>

Resumes a paused sequence, recalculating dates from pause duration.

cancelSequence(sequenceId, companyId, reason): Promise<void>

Cancels sequence and all scheduled items.

skipStep(itemId, companyId): Promise<void>

Skips a single follow-up step.

rescheduleStep(itemId, newDate, companyId): Promise<void>

Reschedules a single step to a new date.

stopOnReply(sequenceId, companyId): Promise<void>

Auto-stops sequence when customer replies.

stopOnOutcome(sequenceId, outcome, companyId): Promise<void>

Stops sequence on opportunity outcome (accepted/rejected/expired/won/lost).

getDueFollowUps(companyId): Promise<DueFollowUp[]>

Returns all overdue follow-up items across active sequences.

generateFollowUpMessage(item, context): Promise<string>

AI-generated contextual message via NIM. References specific product, quantity, quote value, and original inquiry. Falls back to template-based messages if NIM is unavailable.

src/lib/email.ts — Email Sending

Centralised email utility using Resend. Consistent sender config, error handling, and audit logging.

sendEmail(params: SendEmailParams): Promise<SendEmailResult>

Sends an email via Resend API.

- **Input:** { to, subject, html, text?, from?, replyTo?, cc?, bcc?, attachments[] }
- **Output:** { success: boolean, id?: string, error?: string }
- **Default sender:** TradeFlow AI <onboarding@resend.dev>
- **Auto-generates** plain-text fallback from HTML

isEmailConfigured(): boolean

Returns true if RESEND_API_KEY is set.

src/lib/attachment-processor.ts — File Processing

Extracts text content from email attachments for the AI extraction pipeline.

validateAttachment(mimeType, fileSize): ValidationResult

Validates MIME type and file size (max 25 MB).

processAttachment(attachment: AttachmentInput): Promise<ExtractionResult>

Main entry: routes to type-specific processor based on MIME type.

Supported formats: - processPDF(buffer, filename) — PDF text extraction via pdfjs-dist - processExcel(buffer, filename) — Excel parsing via xlsx - processCSV(buffer, filename) — CSV parsing with quoted-field support - processDOCX(buffer, filename) — DOCX extraction via mammoth - processImage(buffer, filename) — Placeholder (status: “needs_ocr”)

Output: { text: string, tables: TableData[], status: "completed" | "failed" | "needs_ocr" }

src/lib/language-detect.ts — Language Detection

Character range heuristics for chatbot language detection.

detectLanguage(text): SupportedLanguage

Detects primary language: English (en), Chinese/Mandarin (zh), Cantonese (yue), or Spanish (es). Uses CJK/Latin ratio + marker words.

buildLanguageInstruction(detectedLang): string

Builds XML instruction block for AI system prompt to respond in the detected language.

isMixedLanguage(text): boolean

Detects code-switching (>20% both Latin and CJK characters).

src/lib/tradeflow-knowledge.ts — Static Knowledge Base

Static knowledge base for the website chatbot. Contains product info, pricing, features, competitive positioning, and dashboard usage guide as XML-structured data.

TRADEFLOW_KNOWLEDGE — Large string with role, language rules, product knowledge, dashboard guide, response format, and guardrails.

PRICING_INFO — Starter/Growth/Enterprise tiers with monthly/annual prices in HKD and feature lists.

src/lib/wechat.ts — WeChat Work Integration

Token management, message sending, signature verification, AES encryption/decryption, XML parsing.

getAccessToken(corpId, secret): Promise<string>

Cached WeChat Work access token (refreshes 5 minutes before expiry).

sendWeChatMessage(corpId, agentId, secret, toUser, message): Promise<void>

Sends a text message via WeChat Work API.

verifySignature(token, msgSignature, timestamp, nonce, encrypt): boolean

SHA1 signature verification for incoming webhooks.

decryptMessage(encodingAESKey, encrypted): string

AES-256-CBC decryption of incoming WeChat messages.

encryptMessage(encodingAESKey, msg, corpId): string

AES-256-CBC encryption for outgoing WeChat messages.

parseWeChatXml(xml): Record<string, string>

Zero-dependency XML parser for WeChat message payloads.

src/lib/web-search.ts — Web Search

Web search utility via Exa API with auto-detection of search-worthy queries.

webSearch(query, numResults?): Promise<string>

Searches via Exa, returns formatted results with titles, URLs, and snippets.

needsWebSearch(message): boolean

Pattern-matches messages that would benefit from web search (price lookups, product comparisons, market research).

src/lib/utils.ts — General Utilities

cn(...inputs): string — Tailwind CSS class merger (clsx + twMerge)

formatDate(date): string — Formats date for en-HK locale

formatRelativeTime(date): string — Returns “just now”, “5m ago”, “2h ago”, “3d ago”

4. API Routes — Admin

Inquiries

Route	Method	Description
/api/admin/inquiries	GET	List all inquiries with filtering, sorting, and pagination
/api/admin/inquiries/[id]	GET, PUT, DELETE	Get, update, or delete a specific inquiry
/api/admin/inquiries/[id]/extract	POST	Run AI extraction on inquiry email text + attachments
/api/admin/inquiries/[id]/attachments	GET, POST	List or upload attachments for an inquiry
/api/admin/inquiries/webhook	POST	Resend inbound email webhook — creates inquiry from received email

Opportunities

Route	Method	Description
/api/admin/opportunities	GET, POST	List or create opportunities
/api/admin/opportunities/[id]	GET, PUT, DELETE	Get, update, or delete an opportunity
/api/admin/opportunities/[id]/stage	PUT	Move opportunity to a new pipeline stage
/api/admin/opportunities/[id]/convert	POST	Convert opportunity to quote

Quotes

Route	Method	Description
/api/admin/quotes	GET, POST	List or create quotes
/api/admin/quotes/[id]	GET, PUT, DELETE	Get, update, or delete a quote
/api/admin/quotes/[id]/send	POST	Send quote to customer — generates HTML+PDF, sends via Resend, creates follow-up sequence
/api/admin/quotes/[id]/approve	POST	Approve a quote (status gate before sending)
/api/admin/quotes/[id]/line-items	GET, POST, PUT, DELETE	Manage quote line items
/api/admin/quotes/[id]/version	POST	Create a version snapshot of the quote

Supplier RFQs

Route	Method	Description
/api/admin/rfqs	GET, POST	List or create supplier RFQs
/api/admin/rfqs/[id]	GET, PUT, DELETE	Get, update, or delete an RFQ
/api/admin/rfqs/[id]/send	POST	Send RFQ to supplier — generates branded email, sends via Resend

Supplier Quotes

Route	Method	Description
/api/admin/supplier-quotes	GET, POST	List or create supplier quotes
/api/admin/supplier-quotes/[id]	GET, PUT, DELETE	Get, update, or delete a supplier quote
/api/admin/supplier-quotes/compare	GET	Side-by-side comparison of supplier quotes

Suppliers

Route	Method	Description
/api/admin/suppliers	GET, POST	List or create suppliers
/api/admin/suppliers/[id]	GET, PUT, DELETE	Get, update, or delete a supplier
/api/admin/suppliers/[id]/quotes	GET	List all quotes from a specific supplier
/api/admin/suppliers/[id]/rfqs	GET	List all RFQs sent to a specific supplier

Customers & Contacts

Route	Method	Description
/api/admin/customers	GET, POST	List or create customers
/api/admin/customers/[id]	GET, PUT, DELETE	Get, update, or delete a customer
/api/admin/contacts	GET, POST	List or create contacts
/api/admin/contacts/[id]	GET, PUT, DELETE	Get, update, or delete a contact

Follow-ups

Route	Method	Description
/api/admin/follow-ups	GET, POST	List or create follow-up sequences
/api/admin/follow-ups/[id]	GET, PUT, DELETE	Get, update, or delete a sequence
/api/admin/follow-ups/[id]/items/[itemId]	PUT, DELETE	Update or delete a specific follow-up step

Products

Route	Method	Description
/api/admin/products	GET, POST	List or create products

Knowledge Base & FAQ

Route	Method	Description
/api/admin/knowledge	GET, POST	List or create knowledge base articles
/api/admin/knowledge/scrape	POST	Scrape a URL and add content to knowledge base (uses Exa)
/api/admin/faq	GET, POST	List or create FAQ rules

Documents

Route	Method	Description
/api/admin/documents	GET, POST	List or upload documents (Supabase Storage)
/api/admin/documents/[id]	GET, DELETE	Get or delete a document

Conversations

Route	Method	Description
/api/admin/conversations	GET	List all conversations with contact info
/api/admin/conversations/[id]/messages	GET	List messages in a conversation

Settings & Config

Route	Method	Description
/api/admin/settings	GET, PUT	Get or update company settings (system prompt, integrations, email config)
/api/admin/company	GET, PUT	Get or update company profile
/api/admin/goals	GET, PUT	Get or update sales goals
/api/admin/email/test	POST	Send a test email via Resend
/api/admin/confirm-email	POST	Confirm email address
/api/admin/pending-companies	GET	List companies pending approval

WhatsApp

Route	Method	Description
/api/admin/whatsapp/embedded-signup	POST	WhatsApp Business embedded signup flow

5. API Routes — Billing

Route	Method	Description
/api/billing/checkout	POST	Creates Stripe Checkout session for subscription (Starter HK\$880 / Growth HK\$2,480 / Enterprise HK\$4,880)
/api/billing/portal	POST	Creates Stripe Billing Portal session for managing existing subscription

Checkout Flow

1. User clicks “Subscribe” on /admin/billing
2. Route creates/reuses Stripe customer
3. Creates Checkout Session with correct tier + interval
4. Returns Stripe URL → user redirected to Stripe
5. On success, Stripe webhook fires `checkout.session.completed`
6. Webhook updates `companies.subscription_status` to active

6. API Routes — Chat & AI

Route	Method	Description
/api/chat	POST	AI chat endpoint — processes messages through NIM LLM with knowledge base, FAQ matching, language detection, and web search fallback

Chat Flow

1. Receive user message + conversation ID
2. Load company context (products, FAQ rules, knowledge base)
3. Detect language
4. Match FAQ patterns (exact match first)
5. Build system prompt with company data
6. Call NVIDIA NIM API (`meta/llama-3.1-8b-instruct`)
7. Save user message + AI response to database
8. Return reply + metadata

7. API Routes — Cron Jobs

Route	Method	Auth	Schedule	Description
/api/cron/follow-ups	GET	CRON_SECRET	Every 15 min	Processes due follow-up items — generates AI messages and sends via email
/api/cron/workflow	GET	CRON_SECRET	Every 10 min	Processes pending workflow jobs (RFQ emails, etc.)

Follow-up Cron Flow

1. Query `follow_up_items` where `status=scheduled` AND `scheduled_for <= now`
2. Filter to active sequences only
3. For each item: build context → generate message via NIM → send via Resend
4. Mark items as sent or failed
5. Log audit events

Workflow Cron Flow

1. Query `workflow_jobs` where `status=pending`
2. Mark as processing
3. Execute job (currently: `send_rfq_email`)
4. Mark as completed or failed

Vercel Cron Config (`vercel.json`)

```
{
  "crons": [
    { "path": "/api/cron/follow-ups", "schedule": "0,15,30,45 * * * *" },
    { "path": "/api/cron/workflow", "schedule": "0,10,20,30,40,50 * * * *" }
  ]
}
```

8. API Routes — Webhooks

Route	Method	Description
<code>/api/webhooks/whatsapp</code>	GET, POST	WhatsApp Cloud API webhook — GET handles Meta verification, POST processes inbound messages with dedup, rate limiting, and subscription checks
<code>/api/webhooks/wechat</code>	GET, POST	WeChat Work webhook — GET handles server verification, POST processes incoming messages
<code>/api/webhooks/stripe</code>	POST	Stripe webhook — processes subscription lifecycle events

WhatsApp Webhook

- **GET:** Meta verification (`hub.mode`, `hub.verify_token`, `hub.challenge`)
- **POST:** Inbound message processing
 - 15-minute dedup window
 - Rate limits to 30 messages/minute
 - Checks subscription status (active vs expired)
 - Supports keyword overrides: “stop” (opt-out), “human” (handoff), “start” (opt-in)
 - Routes to `handleChat()` for AI response

Stripe Webhook

Handles these events: - `checkout.session.completed` — Creates subscription, updates company - `customer.subscription.updated` — Syncs subscription status - `customer.subscription.deleted` — Downgrades to free - `invoice.payment_failed` — Marks as `past_due`

9. API Routes — Public

Route	Method	Auth	Description
/api/demo-request	POST	No	Submits demo request form — stores in DB + sends admin notification email via Resend
/api/health	GET	No	Health check — pings Supabase database to verify connectivity
/api/research	POST	No	Research endpoint — scrapes prospect website via Exa, analyzes with NIM to generate business development responses

Research Output

Returns: company size, revenue range, headquarters, tech stack, competitors, AI analysis, key insights, personalized approach, email template, and follow-up sequence.

10. Type System

Core Enums

Type	Values
ProcessingStatus	RECEIVED → PARSING → EXTRACTING → NEEDS_REVIEW → READY_FOR_RFQ → QUOTE_DRAFTED → FAILED
InquiryPriority	low, normal, high, urgent
SourceChannel	email, whatsapp, wechat, manual, web_form, phone, file_upload
OpportunityStage	NEW → NEEDS_INFORMATION → QUALIFIED → SOURCING → QUOTE_DRAFT → PENDING_APPROVAL → SENT → NEGOTIATING → WON / LOST / EXPIRED
QuoteStatus	DRAFT → IN_REVIEW → APPROVED → SENT → OPENED → CUSTOMER_REPLIED → ACCEPTED / REJECTED / EXPIRED / SUPERSEDED
SupplierRfqStatus	DRAFT → READY → SENT → PARTIALLY_RESPONDED → COMPLETE / EXPIRED / CANCELLED
FollowUpStatus	scheduled, sent, cancelled, completed
FieldStatus	CONFIRMED, EXTRACTED, INFERRED, MISSING, CONFLICTING

Domain Models

- **Customer** — id, name, email, company, phone, tags, notes, metadata

- **Contact** — id, customer_id, name, email, phone, role, is_primary
- **Supplier** — id, legal_name, trading_name, contact_email, contact_name, is_approved, rating, specializations
- **Inquiry** — id, company_id, subject, body, source_channel, status, priority, detected_language
- **Opportunity** — id, company_id, title, customer_id, stage, estimated_value, currency, product_name
- **SupplierRfq** — id, company_id, opportunity_id, supplier_id, status, subject, message_body, response_deadline
- **SupplierQuote** — id, rfq_id, supplier_id, total_price, currency, payment_terms, lead_time_days
- **Quote** — id, opportunity_id, status, version, currency, valid_until, payment_terms, total_amount, margin, margin_percent
- **QuoteLineItem** — id, quote_id, product_name, quantity, unit, unit_price, total_price, margin, margin_percent
- **FollowUpSequence** — id, opportunity_id, quote_id, status, channel
- **FollowUpItem** — id, sequence_id, step_number, delay_days, scheduled_for, status, message_type, subject, message_body
- **WorkflowJob** — id, company_id, job_type, entity_type, entity_id, status, payload

11. Environment Variables

Required

Variable	Purpose
NEXT_PUBLIC_SUPABASE_URL	Supabase project URL
NEXT_PUBLIC_SUPABASE_ANON_KEY	Supabase anonymous key
SUPABASE_SERVICE_ROLE_KEY	Supabase service role key (server-side)
NIM_API_KEY	NVIDIA NIM API key for AI chat/extraction
RESEND_API_KEY	Resend API key for email sending
NEXT_PUBLIC_APP_URL	App base URL (e.g., https://tradeflow-ai-rho.vercel.app)

Optional

Variable	Purpose
STRIPE_SECRET_KEY	Stripe secret key for billing
STRIPE_WEBHOOK_SECRET	Stripe webhook signing secret
WHATSAPP_ACCESS_TOKEN	WhatsApp Cloud API access token
WHATSAPP_VERIFY_TOKEN	WhatsApp webhook verification token
WECHAT_WORK_CORP_ID	WeChat Work corporation ID
WECHAT_WORK_SECRET	WeChat Work application secret
WECHAT_WORK_TOKEN	WeChat Work callback token
WECHAT_WORK_ENCODING_AES_KEY	WeChat Work AES encryption key
EXA_API_KEY	Exa API key for web search/scraping
SENTRY_DSN	Sentry error tracking DSN
CRON_SECRET	Bearer token for cron job authentication
